

## COURSE MATERIAL

|                                         |                                                                |
|-----------------------------------------|----------------------------------------------------------------|
| <b>SUBJECT</b>                          | <b>BASIC CIVIL AND MECHANICAL<br/>ENGINEERING (ME23AES101)</b> |
| <b>UNIT</b>                             | <b>1</b>                                                       |
| <b>COURSE</b>                           | <b>B.TECH</b>                                                  |
| <b>DEPARTMENT</b>                       | <b>CIVIL</b>                                                   |
| <b>SEMESTER</b>                         | <b>II-I or I-II</b>                                            |
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**1. Course Objectives**

The objectives of this course is to

1. Get familiarized with basic Construction Materials; the scope and importance of Civil Engineering specializations.
2. Introduce the preliminary concepts of Structural and Geotechnical Engineering.
3. Acquire preliminary knowledge on Surveying and Transportation Engineering.
4. Get familiarized with the importance of Water Resources and Environmental Engineering.

**2. Prerequisites**

Students should have knowledge on

1. Basic Mathematics
2. Basic Physics

**3. Syllabus****UNIT I**

Basics of Civil Engineering: Role of Civil Engineers in Society- Various Disciplines of Civil Engineering- Structural Engineering- Geo-technical Engineering- Transportation Engineering - Hydraulics and Water Resources Engineering - Environmental Engineering - Scope of each discipline.  
Construction Materials- Cement-Sand-Aggregate - Bricks- Cement concrete- Steel-Timber.

**4. Course outcomes**

1. Understand various sub-divisions of Civil Engineering and to appreciate their role in ensuring better society and the basic characteristics of Construction Materials.
2. Gain knowledge regarding Structural and Geotechnical Engineering.
3. Explain the concepts of surveying and Transportation Engineering, Water Resources and Environmental Engineering.

**5. Co-PO / PSO Mapping**

| Machine Tools | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | P10 | PO11 | PO12 | PSO1 | PSO2 |
|---------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|------|
| CO1           | 2   | 2   |     |     |     |     |     |     |     |     |      |      | 2    | 2    |
| CO2           | 2   | 2   |     |     |     |     |     |     |     |     |      |      | 2    | 2    |
| CO3           | 2   | 2   |     |     |     |     |     |     |     |     |      |      | 2    | 2    |

## 6. Lesson Plan

| Lecture No. | Weeks | Topics to be covered                                                                               | References |
|-------------|-------|----------------------------------------------------------------------------------------------------|------------|
| 1           | 1     | Basics of Civil Engineering: Role of Civil Engineers in Society                                    | T1         |
| 2           |       | Various Disciplines of Civil Engineering- Structural Engineering                                   | T1, R1     |
| 3           |       | Geo-technical Engineering- Transportation Engineering                                              | T1, R1     |
| 4           |       | Hydraulics and Water Resources Engineering - Environmental Engineering - Scope of each discipline. | T1, R1     |
| 5           | 2     | Construction Materials- Cement                                                                     | T1, R2     |
| 6           |       | Sand-Aggregate                                                                                     | T1, R1     |
| 7           |       | Bricks- Cement concrete                                                                            | T1, R1     |
| 8           |       | Steel-Timber                                                                                       | T1, R1     |
| 9           | 3     | Discussion of objective type questions & Short answer questions                                    | T1, R1     |
| 10          |       | Discussion of Previous year university questions in question papers                                | T1, R1     |

## 7. Activity Based Learning

1. Students can conduct experiments to test the compressive strength, tensile strength, and workability of concrete.
2. Students can investigate the properties of steel, such as its strength, ductility, and corrosion resistance.
3. Students can use surveying equipment, such as tapes or electronic distance measuring instruments, to measure distances.
4. Students can conduct experiments to test the quality of water, including parameters such as pH, turbidity, and dissolved oxygen.
5. Students can investigate the factors that affect traffic flow.

**8. Lecture Notes****1.1 INTRODUCTION****ENGINEERING**

- Engineering is the profession of application of scientific and mathematical knowledge and rational thinking to improve living standards.
- Engineer provides comfort to mankind and makes life comfortable.
- Engineer cleverly uses the available natural resources for the benefit of mankind.
- Thus, engineering is primarily a creative activity aimed at satisfying human needs.
- Engineering and civilization are closely related and each stimulates and advances the other.

**CIVIL ENGINEERING**

- Civil Engineering is the oldest branch of engineering which is growing right from the stone stage civilization.
- Civil engineering is the application of physical and scientific principles.
- An infrastructure development of nation plays a significant role in its economic growth.
- A fast-growing economy warrants an even faster development of infrastructure.
- Civil engineers build the world 's infrastructure.
- Civil engineers shape the progress of nation around the world.
- Civil engineers provide safe and comfort living environment to the public.
- Civil engineers are considered to be indispensable professionals.
- Civil engineers have one of the greatest responsibilities on this planet that is to build the quality of life.
- Civil Engineers develop and improve the services and facilities we use every day.
- From supplying clean water to homes, to processing and recycling waste, to find solutions to problems like pollution.
- Civil engineers design and build bridges, roads, railways and tunnels.
- A civil engineer has the responsibility to design safe structures that meet standard codes in the regions where they are being built.
- They must ensure these structures are designed with efficiency, safety, maintenance and durability in mind.

**CIVIL ENGINEERS CONTRIBUTIONS TO THE WELFARE OF SOCIETY**

- Good planning of towns and extension areas in the cities.
- Assured water supply.
- A good drainage system.
- Pollution free environmental conditions.

- A well planned and built networks of roads and road crossings.
- Railways connections to all important cities and towns.
- Airports and harbours of national and international standards.

## VARIOUS DISCIPLINES OF CIVIL ENGINEERING

Civil Engineering may be divided into the following fields :

- 1) Structural Engineering
- 2) Geo-technical Engineering
- 3) Transportation Engineering
- 4) Hydraulics and Water Resources Engineering
- 5) Environmental Engineering

### 1) STRUCTURAL ENGINEERING

It deals with the structural design and analysis of buildings, bridges etc., taking the loads into consideration and designing the various components of a structure for it's stability.

### 2) GEOTECHNICAL ENGINEERING

- It deals with engineering behaviour of soils and rocks and design of foundation.
- Knowledge from the fields of Geology, Material science and testing etc. are applied by Geotechnical Engineers for safe and economical design of foundations etc.

### 3) TRANSPORTATION ENGINEERING

- It deals with the construction of roads including highways, rails, bridges, tunnels, etc.
- The development of a Nation mainly depends on the communication facilities.
- The Nation wealth is measured in terms of the road and railway facilities available.
- There are three modes of transportation, viz., land, water and air.

### 4) HYDRAULICS AND WATER RESOURCES ENGINEERING

#### (A) HYDRAULICS

- Water is an important need for all living beings.
- Hydraulics deals with mechanics of water flow.
- Study of mechanics of fluid and it's flow characteristics is another important field in Civil Engineering and it is known as hydraulics.
- The problems man encountered in the fields of water supply, irrigation, water power resulted in the development of this branch.

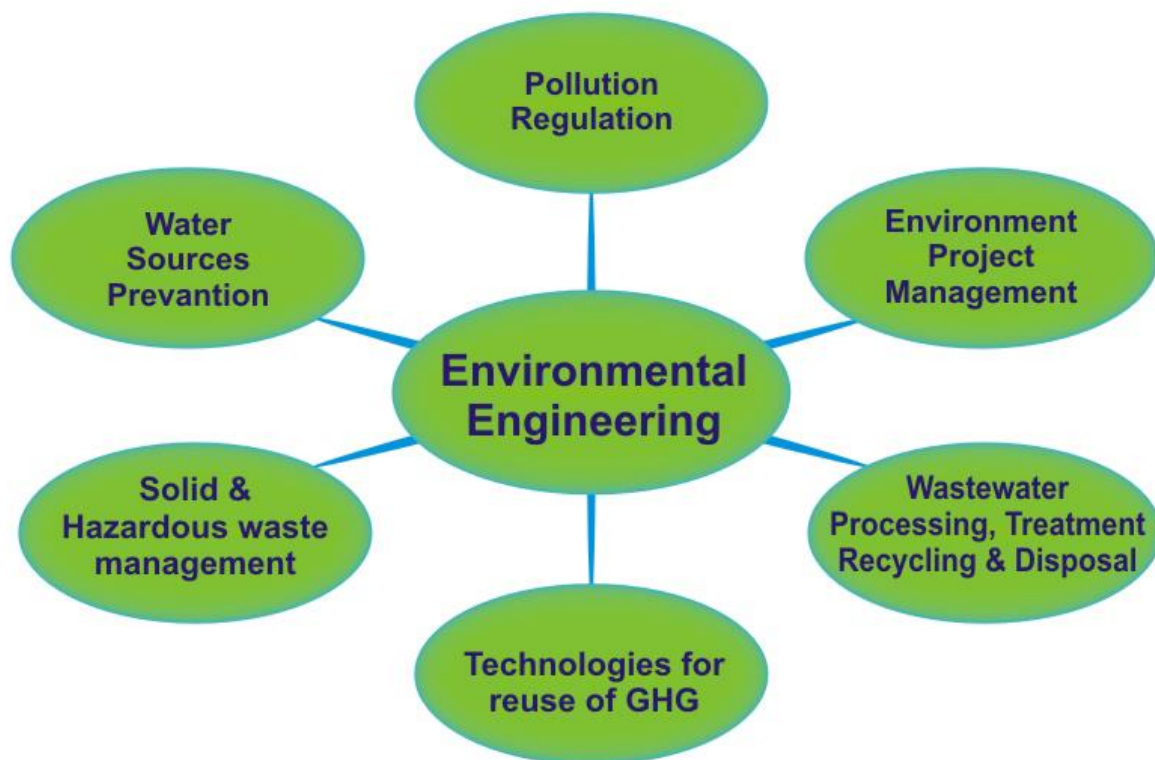
#### (B) WATER RESOURCES ENGINEERING

It deals with the construction and maintenance of canals, reservoirs, dams, distribution channels to the cultivable land, docks, harbors, etc.



## 5) ENVIRONMENTAL ENGINEERING

- Environmental Engineering is the integration of science and engineering principles to improve the natural environment, to provide healthy water, air and land for human habitation and for other organisms, and to remediate pollution sites.
- It deals with the purification of water drinking water supply, disposal of liquid and solid wastes, pollution control etc.
- The Civil Engineer plays an active role in the preservation of the Ecology and environment.



## SCOPE OF VARIOUS DISCIPLINES OF CIVIL ENGINEERING

### 1) STRUCTURAL ENGINEERING

- Structure is the assemblage of two or more basic elements such as beam, slab, column, truss, frame etc.
- Structural engineering is the field of engineering that deals with the structural integrity and strength of a building or a structure.
- Structural engineering is a speciality of civil engineering that ensures the structures are safe, stable and don't collapse under applied loads.
- Structural engineering is mainly focused on analysis and design of the structure.

### STRUCTURAL ANALYSIS

- Loads acting on a structure is ultimately transferred to ground. In doing so, various components of the structure are subjected to internal stresses.
- For example, in a building, load acting on a slab is transferred by slab to ground through beams. Columns and footings.



- Assessing the internal stresses in the components of a structure is known as structural analysis.
- Analysis of the structure according to the principles of structural engineering will make sure that the structure depends on all the necessary design codes.

### **DESIGN OF THE STRUCTURE**

- After analyzing the structure, finding the suitable size of the structural component is known as design of the structure.
- Structures have to be designed so that they can withstand their own weight as well as the loads and pressures that will be placed upon them.
- Structural engineers take steps crucial information about the foundations, roof types, load types, beams, columns, material quality, retaining wall etc.
- A civil engineer has not only to give a safe and elegant structure but he has to give economical sections.

### **RESPONSIBILITIES OF A STRUCTURAL ENGINEER**

- A structural engineer also plays a major role as a team among other professionals like surveyor and architect engineers.
- The following tasks must be performed by a structural engineer
  - Design models of structures using software
  - Assessing the reaction of structures to pressures and stresses.
  - Finalizing the appropriate concrete materials that would be suitable for the structure.
  - Assessing the budget of the project.

### **SKILLS REQUIRED TO BECOME A STRUCTURAL ENGINEER**

It's necessary for every structural engineer to possess the following skills

- Analytical skills
- Detailed orientation
- Creativity
- Strong interpersonal skills
- Communication skills
- Knowledge of construction management
- Up-to-date technical skills

## **2) GEOTECHNICAL ENGINEERING**

- Soil property changes from place to place. Even in the same place it may not be uniform at various depths.
- The soil property may vary from season to season due to variation in moisture content.
- The load from the structure is to be safely transferred to soil. For this, safe bearing capacity of the soil is to be properly assessed.

- The study of soil strata and its different properties is called Geotechnical Engineering.
- Apart from finding safe bearing capacity for foundation of buildings, geotechnical engineering involves various studies required for the design of pavements, tunnels, earthen dams, canals and earth retaining structures.
- Geotechnical Engineers are responsible for evaluating subsurface and soil conditions and materials, using the principles of soil and rock mechanics.
- They are commonly appointed as consultants on construction projects.
- Geotechnical engineers can be involved in ground improvement, in which soil is treated through a variety of different techniques to improve strength and stiffness.

### **3) TRANSPORTATION ENGINEERING**

- Transportation facility is another important need to the public.
- Providing good and economical transportation facilities is an important duty of civil engineers.
- Transportation is an important branch of civil engineering that oversees the planning, designing, construction, implementation, maintenance and operations of various modes of transportation.
- Any product or system that moves people and goods from one place to another falls under the scope of transportation engineering.

#### **MODES OF TRANSPORTATION**

There are three modes of transportation, they are

- i) Land ways
- ii) Water ways
- iii) Air ways

##### **i) Land ways**

- Land transport covers all land-based transportation systems that provide for the movement of people, goods and services.
- Land transport plays a vital role in linking communities to each other.
- Land transport is a key factor in urban planning.
- It consists of two types, they are roadways and railways.

##### **ROADWAYS**

- A road is an identifiable route of travel, usually surfaced with gravel, asphalt or concrete and supporting land passage by foot or by a number of vehicles.
- Providing good and economical roads is an important duty of civil engineers.
- It involves design of base courses, suitable surface finishes, cross drainage works, road intersections, culverts, bridges, tunnels etc.
- For proper planning of this transportation facility, traffic survey is to be carried out.
- So, it involves carrying out traffic survey, design, construction and maintenance of roads.

**RAILWAYS**

- Railway is another important long way transport facility.
- Railway transportation means conveyance of passengers and goods by way of wheeled vehicles running on rail track.
- Design, construction and maintenance of railway lines, signal system are part of transportation engineering.

**ii) WATER WAYS**

- Transportation by water is the slowest when compared to other modes of transportation.
- Water transportation is the process of transport that a watercraft, such as boat, ship or sail boat makes over a body of water, such as a sea, ocean, lake, canal or river.
- If a boat or a ship can successfully pass through a waterway it is known as navigable waterway.
- The transportation by water is possible between the ports on the sea routes or along the rivers or canals where inland transportation facilities are available.

**iii) AIRWAYS**

- The transportation by air is the fastest among all modes of transportation.
- Air travel also provides more comfort apart from saving in transportation time for passengers and the goods between the airports.

**4) HYDRAULICS AND WATER RESOURCES ENGINEERING****A) HYDRAULICS**

- Water is an important need for all living beings.
- Study of mechanics of fluid and its flow characteristics is another important field in Civil Engineering and it is known as hydraulics.
- Hydraulic engineering is the application of the principles of fluid mechanics to problems dealing with the collection, storage, control, transport, regulation, measurement and use of water.
- Before beginning a hydraulic engineering project, one must figure out how much water is involved.
- The hydraulic engineering is intimately related to the design of bridges, dams, channels, canals.
- The hydraulic engineer actually develops conceptual designs for the various features which interact with water such as spillways and outlet works for dams, culverts for highways, canals and related structures for irrigation projects.

**B) WATER RESOURCES ENGINEERING**

- Water is essential to our lives.
- Water is to be supplied for drinking purposes and agricultural purposes.
- Hence, suitable water resources are to be identified and water is to be stored.
- Water resources engineers deal with the physical control of water.

- They work with others to prevent floods, supply water for cities, industry and agriculture.
- Identifying, planning and building water retaining structures like tanks and dams is known as water resources engineering.
- The main responsibility of a water resource engineer is to manage a population's water use and ensure that the treatment of water is safe for human consumption.
- They are also responsible for designing treatment plants, supply systems, pipelines and pump systems to effectively manage wastewater for both private and public consumption.
- Wastewater, if not treated and managed correctly, can have negative health effects on the environment and a community.

## **5) ENVIRONMENTAL ENGINEERING**

- Environmental engineering is the application of science and engineering principles that provide a better environment to give healthy water, air and land for human habitation and other living beings.
- Proper distribution of water to rural areas, towns and cities and disposal of wastewater and solid waste are related to environmental engineering.
- The skills of environmental engineers have become increasingly important as we protect our fragile resources.
- Environmental engineers translate physical, chemical and biological processes into systems to destroy toxic substances, remove pollutants from water, reduce hazardous waste volumes, eliminate contaminants from the air and develop water supplies.
- Environmental engineers are called upon to resolve the problems of providing safe drinking water, cleaning up contaminated sites with hazardous materials, disposing of wastewater and managing solid wastes.
- Environmental engineering while tackling all these problems provides healthy environment to public.

## **CONSTRUCTION MATERIALS**

### **INTRODUCTION**

- Several materials are required in construction.
- The materials used in the construction of engineering structures such as buildings, bridges and roads are called construction materials or building materials or engineering materials.
- They include cement, sand, aggregates, bricks, cement concrete, steel and timber

### **CEMENT**

- Cement is commonly used binding material in construction.

- The cement is obtained by burning a mixture of calcareous (calcium) and argillaceous (clay) material at a very high temperature and then grinding the clinker so produced to a fine powder.
- It was first produced by a mason Joseph Aspdin in England in 1824.
- He patented it as cement.

### COMPOSITION OF ORDINARY CEMENT

- The ordinary cement contains two basic ingredients namely argillaceous and calcareous.
- In argillaceous materials, the clay predominates and in calcareous materials, the calcium carbonate predominates.
- A typical chemical analysis of a good ordinary cement is as follows
  - Lime ( $\text{CaO}$ ) – 62 %
  - Silica ( $\text{SiO}_2$ ) – 22 %
  - Alumina ( $\text{Al}_2\text{O}_3$ ) – 5 %
  - Calcium sulphate – 4 %
  - Iron oxide – 3 %
  - Magnesia – 2%
  - Sulphur – 1 %
  - Alkalies – 1 %

### HYDRATION OF CEMENT

- Hydration is the chemical reaction between cement and water by which Portland cement becomes a firm, hardened mass.
- The hydrated cement is relatively insoluble in water.
- During hydration, heat is evolved and temperatures in mass concrete may reach 60 °C.

### HEAT OF HYDRATION OF CEMENT

The heat of hydration is the heat generated when water and Portland cement react.

### PROPERTIES OF GOOD CEMENT

Following are the important properties of good cement

- It should act as good binding material.
- It should be easily workable.
- It should offer good resistance to moisture.
- It should possess good plasticity.
- It should harden early.
- It should impart strength to masonry.



## PROPERTIES OF CEMENT

### 1) CHEMICAL PROPERTIES

Portland cement consists of the following chemical compounds

- 1) Tricalcium Silicate -  $3CaO.SiO_2$  - 40%
  - 2) Dicalcium Silicate ( $C_2S$ ) -  $2CaO.SiO_2$  - 30%
  - 3) Tricalcium Aluminate ( $C_3A$ )  $3CaO.Al_2O_3$  - 11 %
  - 4) Tetra-calcium Alumino Ferrite ( $C_4AF$ ) -  $4CaO.Al_2O_3Fe_2O_3$  - 11%
- There may be small quantities of impurities present such as calcium oxide (CaO) and magnesium oxide (MgO).
  - When water is added to cement,  $C_3A$  is the first to react and cause initial set and it generates great amount of heat.
  - $C_3S$  hydrates early and develops strength in the first 28 days and it also generates heat.
  - $C_2S$  is the next to hydrate. It hydrates slowly and is responsible for increase in ultimate strength.
  - $C_4AF$  is comparatively inactive compound.

### 2) PHYSICAL PROPERTIES

The following physical properties should be checked before selecting a Portland cement for the civil engineering works.

- a) Fineness
- b) Setting time
- c) Soundness
- d) Crushing strength

#### a) Fineness

- It is measured in terms of percentage of weight retained after sieving the cement through 90-micron sieve.
- According to IS code specification weight retained on the sieve should not be more than 10 per cent.

#### (b) Setting time

- A period of 30 minutes as minimum setting time for initial setting and a maximum period of 600 minutes as maximum setting time is specified by IS code.

#### Initial Setting time of Cement

Initial setting time is regarded as the time elapsed between the moments that the water is added to the cement, to the time that the paste starts losing its plasticity.

**Final Setting time of Cement**

The final setting time is the time elapsed between the moment the water is added to the cement, and the time when the paste has completely lost its plasticity and has attained sufficient firmness to resist certain definite pressure.

**(c) Soundness**

- Once the concrete has hardened it is necessary to ensure that no volumetric changes takes place.
- The cement is said to be unsound, if it exhibits volumetric instability after hardening.
- IS code recommends test with Le Chatelier mould for testing this property.
- At the end of the test, the indicator of Le Chatelier mould should not expand by more than 10 mm.

**(d) Crushing strength**

- For this mortar cubes are made with standard sand and tested in compression testing machine as per the specification of IS code.
- The minimum strength specified is  $16 \text{ N/mm}^2$  after 3 days and  $22 \text{ N/mm}^2$  after 7 days of curing.

**TYPES OF CEMENT**

In addition to Ordinary Portland Cement, there are many varieties of cement. They are

- 1) White Cement
- 2) Coloured Cement
- 3) Quick Setting Cement
- 4) Rapid Hardening Cement
- 5) Low Heat Cement
- 6) Pozzolana Cement
- 7) Expanding Cement
- 8) High Alumina Cement
- 9) Blast Furnace Cement
- 10) Acid Resistant Cement
- 11) Sulphate Resisting Cement
- 12) Fly ash Blended Cement

**1) WHITE CEMENT**

- This is just a variety of ordinary cement and it is prepared from such raw materials which are practically free from colouring oxides of iron, manganese or chromium.
- For burning of this cement, the oil fuel is used instead of coal.
- White cement is used for floor finishes, plastering works, fixing tiles etc.
- The first white cement factory was setup in Kottayam, Kerala in 1956.



- White cement is costlier than ordinary cement because of specific requirements imposed upon the raw materials and the manufacturing process.

## **2) COLOURED CEMENT**

- The cements of desired colours are produced by mixing mineral pigments with ordinary cement.
- Chromium oxide gives green colour.
- Cobalt produces blue colour.
- Iron oxide is added to give brown, red and yellow colour.
- Manganese dioxide gives black colour to the cement.
- These cements are used for giving finishing of floors, walls, roofs etc.

## **3) QUICK SETTING CEMENT**

- This cement is produced by adding a small percentage of aluminium sulphate and by finely grinding the cement.
- The percentage of gypsum is also greatly reduced.
- This cement starts setting within 5 minutes after addition of water and it becomes hard like stone in less than 30 minutes.
- This cement is used to lay concrete under static or slowly running water.

## **4) RAPID HARDENING CEMENT**

- By burning at high temperatures, increasing lime content and grinding very fine, rapid hardening cement is produced.
- The initial and final setting time of this cement is same as ordinary cement.
- But it attains high strength in early days.
- This cement is slightly costlier than ordinary cement, but it offers the following advantages
  - As it sets rapidly, the construction work may be carried out speedily.
  - This cement requires short period of curing and permits earlier removal of formwork.
  - It obtains strength in a short period.
  - It is light in weight.
  - It is not damaged easily.
  - The structural members constructed with this cement may be loaded earlier.
  - Use of this cement results in economic design.

## **5) LOW HEAT CEMENT**

- In mass concrete works like construction of dams, the heat produced due to heat of hydration will not get dispersed easily.
- It may give rise to cracks.
- In order to reduce the amount of heat, this type of cement is used.
- It contains lower percentage of tricalcium aluminate of about 5% and higher percentage of dicalcium silicate of about 46%.

- This cement possesses less compressive strength.

## **6) POZZOLANA CEMENT**

- The pozzolana is a volcanic powder.
- It is found in Italy.
- It can also be processed from shales and certain types of rocks.
- The percentage of pozzolana materials should be between 10 to 30 %.
- This cement is used to prepare mass concrete of lean mix and for marine structures.

Following are the advantages of this cement

- It attains compressive strength with age
- It can resist action of sulphates.
- It evolves less heat during setting.
- It imparts higher degree of water tightness.
- It is cheap.
- It offers greater resistance to expansion.
- It possesses higher tensile strength.

Following are the disadvantages of this cement

- It's compressive strength in early days is less.
- It possesses less resistance to the erosion and weathering action.

## **7) EXPANDING CEMENT**

- This type of cement is produced by adding an expanding medium like sulpho-aluminate and a stabilizing agent to the ordinary cement.
- Hence this cement expands whereas other cements shrink.
- The expanding cement is used for the construction of water retaining structures and also for repairing the damaged concrete surfaces.

## **8) HIGH ALUMINA CEMENT**

- It is manufactured by calcining a mixture of lime and bauxite.
- It is more resistant to attack by sulphate and acid also to the action of frost.
- It contains aluminates between 35 to 55 %.
- It is used for works under water.

Following are the advantages of this cement

- It develops almost full strength within 24 hours of adding water.
- It can withstand high temperature.
- It resists the action of acids in a better way.

Following are the disadvantages of this cement

- It cannot be used in mass construction as it evolves great heat and as it sets soon.
- It is costly.

## **9) BLAST FURNACE CEMENT**

- For this cement, the slag as obtained from blast furnace is used.

- The slag is a waste product in the manufacturing process of pig-iron and it contains the basic elements of cement namely alumina, lime and silica.
- The properties of this cement are more or less same as that of ordinary cement, but it is cheap, since it utilizes waste product.
- This cement is durable but it needs longer period of curing since it gains strength slowly.

#### **10) ACID RESISTANT CEMENT**

- An acid resistant cement is composed of the following
  - (i) Acid resistant aggregates such as quartz, quartzites etc.
  - (ii) Additive such as sodium fluosilicate
  - (iii) Aqueous solution of sodium silicate or soluble glass
- This cement has good resistance to acid and water.
- It is commonly used in the construction of chemical factories.

#### **11) SULPHATE RESISTANT CEMENT**

- By keeping the percentage of tricalcium aluminate below 5%, this cement is produced.
- This cement is similar to OPC except it contains more silicates and less quantity of aluminates.
- This cement is used for under water structures particularly exposed to alkali action.
- Examples of such structures are canals, culverts etc.

#### **12) FLY ASH BLENDED CEMENT**

- Fly ash is a by-product in thermal power stations.
- They contain silica and other chemicals.
- Particles of fly ash are very small and they create air pollution problem.
- Thermal power stations spend a lot of money to arrest fly ash and dispose.
- One of the best way to dispose it is to mix with cement.
- We have to further process the fly ash to make it suitable to blend with cement.
- 20 to 30 % fly ash is used for blending.
- Fly ash blended cement have superior quality of resistance to weather.
- Hence, concrete produced with such cement is durable.
- The ultimate strength gained is same as that with OPC.

#### **GRADES OF CEMENT**

Ordinary Portland Cement (OPC) is classified in to following three grades

- 1) OPC – 33 Grade
- 2) OPC – 43 Grade
- 3) OPC – 53 Grade

The number 33, 43, 53 represents the strength of cement cured for 28 days in N/mm<sup>2</sup>.

| Grade/Curing period | Strength (N/mm <sup>2</sup> ) |    |    |
|---------------------|-------------------------------|----|----|
|                     | 3                             | 7  | 28 |
| <b>33</b>           | 16                            | 22 | 33 |
| <b>43</b>           | 23                            | 33 | 43 |
| <b>53</b>           | 27                            | 37 | 53 |

In the modern construction activities higher grade cements have become so popular that 33 grade cement almost out of the market.

### USES OF CEMENT

Cement is used widely for the construction of various structures.

Some of them are

- Cement slurry is used for filling cracks in concrete structures.
- Cement mortar is used for masonry works, plastering, pointing etc.
- Cement concrete is used for the construction of various structures like buildings, bridges, water tanks, tunnels etc.
- It is useful for the construction of roads, footpaths etc.

### AGGREGATES

- The aggregates are mixed with cement to form concrete or cement mortar.
- Aggregates are generally cheaper than cement and impart greater volume, stability and durability to concrete.
- There are two types of aggregates
  - (i) **Fine aggregate**
    - Fine aggregate usually consists of sand or crushed stone.
  - (ii) **Coarse aggregate**
    - Coarse aggregates consists of gravel (pebbles), fragments of broken stone.

### SAND

- Sand is a granular material composed of finely divided mineral particles.
- Sand is a non-renewable resource.
- Sand is suitable for making concrete.
- The most important function of sand is to assist in producing workability and uniformity in mixture.
- The sand also assists the cement paste to hold the coarse aggregate particles in suspension.

### NATURAL SOURCES OF SAND

- The sand particles consists of small grains of silica.
- It is formed by decomposition of sand stones due to various effects of weather.
- According to the natural sources from which the sand obtained, it is of the following three types

- 1) Pit sand
- 2) River sand
- 3) Sea sand

## 1) PIT SAND

- This sand is found as deposits in soil and it is obtained by forming pits in to soils.
- It is excavated from a depth of about 1 m to 2 m from ground level.
- The pit sand consists of sharp angular grains which are free from salts.
- The pit sand is an excellent material for mortar or concrete work.
- Pit sand must be made free from clay and other organic materials before it can be used in mortar.

## 2) RIVER SAND

- This sand is obtained from banks or beds of rivers.
- The river sand consists of fine rounded grains due to mutual attrition under action of water current.
- The colour of river sand is almost white.
- As river sand is usually in clean condition, it is widely used for all purposes.

## 3) SEA SAND

- Sea sand is obtained from the sea shores.
- The sea sand consists of fine rounded grains.
- Sea sand is light brown in colour.
- The sea sand contains salts and these salts attract moisture from the atmosphere.
- Such absorption causes dampness, efflorescence and disintegration of work.
- Sea sand increases the setting time of cement.
- Due to all such reasons, it is the general rule to avoid use of sea sand for all engineering purposes even though it is available in plenty.

## CLASSIFICATION OF SAND

According to the size of the grains, the sand is classified as fine, coarse and gravelly.

### 1) FINE SAND

- The sand passing through a sieve with clear openings of 1.5875 mm is known as fine sand.
- Fine sand is mainly used for plastering.

### 2) COARSE SAND

- The sand passing through a sieve with clear openings of 3.175 mm is known as coarse sand.
- It is generally used for masonry work.



**3) GRAVELLY SAND**

- The sand passing through a sieve with clear openings of 7.62 mm is known as gravelly sand.
- It is generally used for concrete work.

**BULKING OF SAND**

- The increase in the volume of sand due to the presence of moisture is known as bulking of sand.
- This is due to the fact that moisture forms a film of water around the sand particles and this results in an increase in the volume of sand.
- The extent of bulking depends on the grading of sand.
- The finer the material the more will be the increase in volume for the given moisture content.
- For a moisture content of 5 – 8 percent, the increase in volume may be about 20 – 40 percent depending upon the gradation of sand.
- When the moisture content is further increased, the sand particles pack near each other and the amount of bulking is decreased.
- Hence dry sand and the sand completely flooded with water have practically the same volume.
- The volumetric proportioning of sand depends upon the extent of bulking.
- It is more with fine sands than with coarse sands.
- If proper allowances are not made for bulking of sand, the cost of concrete and mortar increases and it results in inadequate sand.
- This makes the mix harsh and difficult for working and placing.

**PROPERTIES OF GOOD SAND**

- It should be clean and coarse.
- It should be free from any organic matter or vegetable matter.
- It should be chemically inert.
- It should contain sharp, angular, coarse and durable grains.
- It should not contain salts which attract moisture from the atmosphere.
- It should be well graded, i.e., it should contain particles of various sizes in suitable proportions.
- It should be strong and durable.

**AGGREGATES**

- Aggregates are the important constituents in concrete.
- Aggregates give body to the concrete, reduce shrinkage and effect economy.
- Aggregates constitute major volume in concrete, they occupy 70-80 percent of the volume of the concrete.
- Aggregates are the most mined materials in the world.
- Aggregates are a component of composite materials such as concrete and asphalt.



- Aggregates serve as reinforcement to add strength to the overall composite material.
- Aggregates are also used as base material under roads, railroads etc.

## **SOURCES OF AGGREGATES**

- Almost all natural aggregates originate from bed rocks.
- There are three kinds of rocks, namely igneous, sedimentary and metamorphic.
- These classifications are based on the mode of formation of rocks.

### **i) IGNEOUS ROCKS**

The rocks which are formed by the cooling of magma are known as the igneous rocks.  
Examples: Granite, dolerite, basalt etc.

### **ii) SEDIMENTARY ROCKS**

These rocks are formed by the deposition of products of weathering on the pre-existing rocks.

Examples: Gravel, sandstone, limestone, gypsum, lignite etc.

### **iii) METAMORPHIC ROCKS**

- These rocks are formed by the change in character of the pre-existing rocks.
- The igneous rocks as well as sedimentary rocks are changed in character when they subjected to great heat and pressure.
- The process of change is known as metamorphism.

Examples: Marble, slate, quartzite etc.

## **AGGREGATES FROM IGNEOUS ROCKS**

Most of the igneous rocks make highly satisfactory concrete aggregates because they are normally hard, tough and dense.

## **AGGREGATES FROM SEDIMENTARY ROCKS**

The quality of aggregates derived from sedimentary rocks will vary in quality depending upon the cementing material and the pressure under which these rocks are originally compacted.

## **AGGREGATES FROM METAMORPHIC ROCKS**

Many metamorphic rocks particularly quartzite and gneiss have been used for production of good concrete aggregates.

## **CLASSIFICATION OF AGGREGATES**

### **1) CLASSIFICATION ACCORDING TO GEOLOGICAL ORIGIN**

#### **A) NATURAL AGGREGATES**

These are generally obtained from natural deposits of sand and gravel or from quarries by cutting rocks.

## **B) ARTIFICIAL AGGREGATES**

The most widely used aggregates are clean broken bricks and air-cooled fresh blast furnace slag.

## **2) CLASSIFICATION ACCORDING TO SIZE**

### **A) FINE AGGREGATE**

The aggregates which passes through a 4.75 mm IS sieve and predominantly retained on the 75  $\mu\text{m}$  sieve are called fine aggregates.

### **B) COARSE AGGREGATE**

The aggregates which are retained on the 4.75 mm IS sieve having particle size more than 4.75 mm are termed as coarse aggregates.

## **3) CLASSIFICATION ACCORDING TO SHAPE**

- The particle shapes of aggregates influence the properties of fresh concrete more than those of hardened concrete.
- Depending upon the particle shape, the aggregate may be classified as follows

- A) Rounded aggregate
- B) Irregular aggregate
- C) Angular aggregate
- D) Flaky and elongated aggregates

### **A) ROUNDED AGGREGATES**

- The rounded aggregates are completely shaped by attrition and available in the form of seashore gravel.
- Rounded aggregates result in the minimum percentage of voids (32-37%) and hence gives more workability.
- They are not considered for high strength concrete because of poor interlocking behaviour and weak bond strength.

### **B) IRREGULAR AGGREGATES**

- The irregular or partly rounded aggregates are partly shaped by attrition and these are available in the form of pit sands and gravel.
- Irregular aggregates may result 35-37% of voids.
- These aggregates will give lesser workability when compared to rounded aggregates.
- The bond strength is slightly higher than rounded aggregates but not as required for high strength concrete.

**C) ANGULAR AGGREGATES**

- Angular aggregates result in maximum percentage of voids (38-45%) and hence gives less workability.
- They give 10-20% more compressive strength due to development of stronger aggregate-mortar bond.
- So, these are useful in high strength concrete manufacturing.

**D) FLAKY AND ELONGATED AGGREGATES**

- When the aggregate thickness is small when compared to it's width and length, it is said to be flaky aggregate.
- When the length of aggregate is larger than the other two dimensions then it is called elongated aggregate.

**4) CLASSIFICATION BASED ON UNIT WEIGHT**

The aggregates can also be classified according to their unit weight as follows

**A) NORMAL WEIGHT AGGREGATE**

- The normal weight aggregates are such as sand, gravel material which we regularly use in construction.
- The normal weight aggregates are used for column, beam and slab.
- The density of normal weight aggregates ranges between 1520 to 1680 Kg/Cum.

**B) LIGHT WEIGHT AGGREGATE**

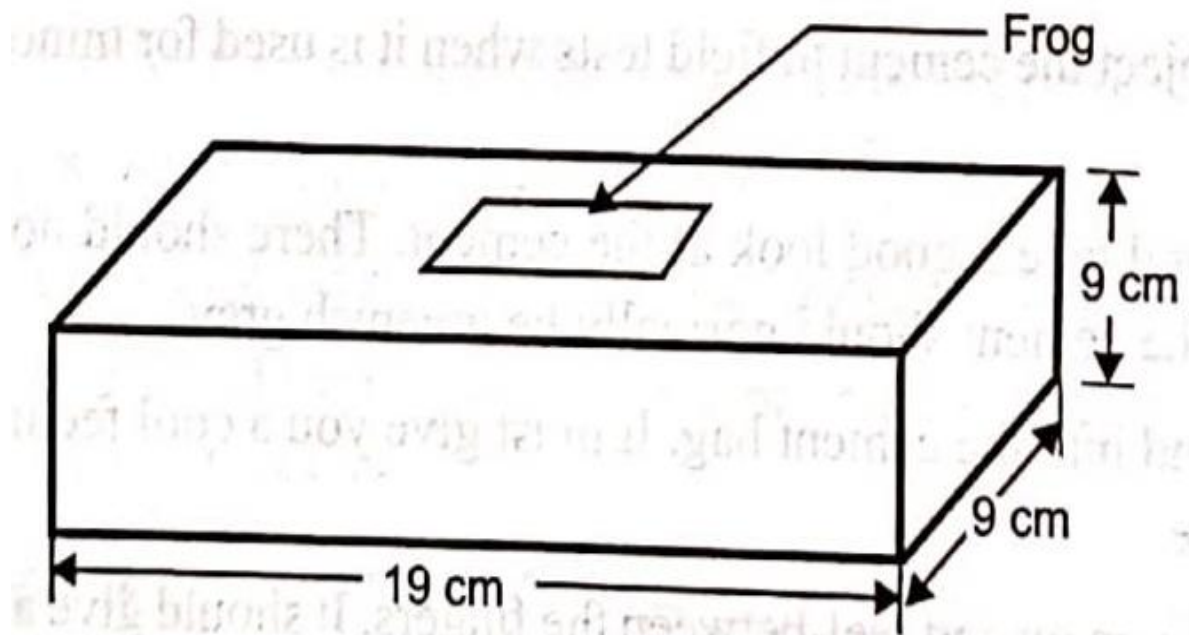
- The density of light weight aggregates ranges between 800 to 1100 Kg/Cum.
- The light weight aggregates will have a high absorption value compared to other aggregates.
- The light weight aggregates are used as a filling material in flooring, deck slab etc.

**C) HEAVY WEIGHT AGGREGATE**

- The density of heavy weight aggregate ranges between 2100 to 2900 Kg/Cum.
- The main drawback of heavy weight aggregate is it produces very low workability in concrete.

**BRICKS**

- Bricks are one of the oldest known building material.
- Other than wood, brick is the most commonly used building material in the construction of buildings.
- Brick is extensively used at present as a leading material of construction because of it's durability, strength, reliability, low cost and easy availability.
- The bricks have been used all over the world in every class and kind of building.
- Bricks are reasonably heat and sound proof.
- According to Indian standards, size of the brick is 19 cm X 9 cm X 9cm.



### COMPOSITION

- Bricks are typically made from clay, which is a natural material abundant in many regions.
- The clay is mixed with water and other additives to improve its properties, such as strength and durability.
- Once the mixture is formed into the desired shape, the bricks are fired in a kiln at high temperatures to harden them.

### OPERATIONS INVOLVED IN THE MANUFACTURE OF BRICKS

Brick making involves the following operations

- 1) Preparation of clay
- 2) Moulding
- 3) Drying
- 4) Burning

### USES OF BRICKS

- Bricks are extensively used as a leading material of construction.
- Bricks are extensively used for construction of load-bearing walls and partition walls.
- A fire brick is used for lining the interiors of ovens, chimneys and furnaces.
- Broken bricks are used as a ballast material for railway tracks and also as a road metal.
- Bricks are also used for face work when artistic effect is required.

### COMPOSITION OF GOOD BRICK EARTH

#### 1) Alumina

- Alumina is the chief constituent of every kind of clay.

- It constitutes about 20% to 30%.
- Alumina imparts plasticity to the clay.
- Excess of alumina makes raw bricks to shrink and warp during drying and burning.

## 2) Silica

- A good brick earth should contain 50% to 60% of silica.
- It prevents cracking, shrinking and warping of raw bricks.
- It gives uniform shape and durability to the bricks.
- Excess of silica makes the brick brittle and useless.

## 3) Lime

- It constitutes 5%.
- The lime prevents shrinkage of raw bricks.
- It enables the silica to melt in burning and bind the particles of brick together.
- Excess of lime makes the brick to melt and the shape will be lost.

## 4) Oxide of Iron

- It constitutes 5% to 6%.
- Oxide of iron enhances the durable quality.
- It imparts red color to the bricks.
- The excess of oxide of iron makes the bricks dark blue or blackish.

## 5) Magnesia

- A small quantity of magnesia in brick earth decreases shrinkage.
- It imparts yellow tint to the bricks.
- Excess of magnesia leads to the decay of bricks.

## 6) Alkalies

- Alkalies in bricks absorb moisture from it.
- They cause the bricks to fuse, twist and warp during burning.

## PROPERTIES OR QUALITIES OF GOOD BRICKS

1. It should have deep red color or cherry red color.
2. It should have perfect and sharp edges.
3. It should be free from cracks or flaws.
4. It should have rectangular shape of standard size 19 cm X 9 cm X 9 cm.
5. It should be burnt equally and evenly.
6. Clear metallic ringing sound should come when struck with each other or with hammer.
7. When broken, it should show homogenous and compact structure and free from voids.
8. It should be sufficiently hard.
9. No impression should be left when scratched with finger nails.
10. It should not break when dropped flat on hard ground from a height of about one meter.



11. When kept immersed in water for 16 hours, it should not absorb water more than 20% of its own weight.
12. It should have low thermal conductivity and high sound insulation.
13. Crushing strength should be in between  $3.5 \text{ N/mm}^2$  to  $7 \text{ N/mm}^2$
14. Average weight of a brick should be in between 3 to 3.5 kg.

## **TYPES OF BRICKS**

- 1) Building bricks
- 2) Paving bricks
- 3) Fire bricks
- 4) Special bricks

### **1) Building Bricks**

These bricks are used for the construction of walls.

### **2) Paving Bricks**

These are vitrified bricks and are used as pavers.

### **3) Fire Bricks**

- These bricks are specially made to withstand furnace temperature.
- Silica bricks belong to this category.

### **4) Special Bricks**

- These bricks are different from the commonly used building bricks with respect to their shape and the purpose for which they are made.
- Some of such bricks are listed below
  - (a) Specially shaped bricks
  - (b) Facing bricks
  - (c) Perforated building bricks
  - (d) Burnt clay hollow bricks
  - (e) Sewer bricks
  - (f) Acid resistant bricks.

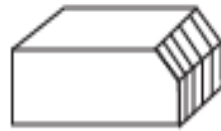
#### **(a) Specially Shaped Bricks**

- Bricks of special shapes are manufactured to meet the requirements of different situations.
- Some of them are shown in below figure.

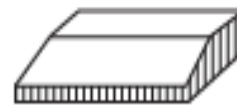




Bull nosed brick



Cant brick



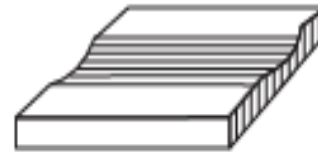
Plinth brick



Coping brick



Cornice brick



Channel brick

**(b) Facing Bricks**

- These bricks are used in the outer face of masonry. Once these bricks are provided, plastering is not required.
- The standard size of these bricks are  $190 \times 90 \times 90$  mm or  $190 \times 90 \times 40$  mm.

**(c) Perforated Building Bricks**

- These bricks are manufactured with area of perforation of 30 to 45 per cent.
- The area of each perforation should not exceed  $500 \text{ mm}^2$
- The perforation should be uniformly distributed over the surface.
- They are manufactured in the size  $190 \times 190 \times 90$  mm and  $290 \times 90 \times 90$  mm.

**(d) Burnt Clay Hollow Bricks**

- They are light in weight.
- They are used for the construction of partition walls.
- They provide good thermal insulation to buildings.
- They are manufactured in the sizes  $190 \times 190 \times 90$  mm,  $290 \times 90 \times 90$  mm and  $290 \times 140 \times 90$  mm.
- The thickness of any shell should not be less than 11 mm and that of any web not less than 8 mm.

**(e) Sewer Bricks**

- These bricks are used for the construction of sewage lines.
- They are manufactured from surface clay, fire clay shale or with the combination of these.
- They are manufactured in the sizes  $190 \times 90 \times 90$  mm and  $190 \times 90 \times 40$  mm.
- The average strength of these bricks should be a minimum of  $17.5 \text{ N/mm}^2$
- The water absorption should not be more than 10 per cent.

## CLASSIFICATION OF BRICKS BASED ON THEIR QUALITY

The bricks used in construction are classified as

- 1) First class bricks
- 2) Second class bricks
- 3) Third class bricks
- 4) Fourth class bricks

### 1) First class bricks

- These bricks are of high quality and are of uniform in size, shape and colour.
- They have sharp edges and are free from cracks and other defects.
- They are burnt in kilns.
- They fulfil all desirable properties of bricks.

#### Uses:

- First class bricks are typically used for load bearing structures and exposed walls.
- These are also used in floors and reinforced bricks slabs.

### 2) Second class bricks

- These bricks are ground moulded and burnt in kilns.
- The edges may not be sharp and uniform.
- The shape of the bricks is also uneven.
- The surface may be somewhat rough.
- These bricks have strength and durability.

#### Uses:

- These are used for internal walls that are not exposed to atmosphere.

### 3) Third class bricks

- These bricks are ground moulded and burnt in clamps.
- Their edges are somewhat distorted.
- They produce dull sound when struck together.
- These bricks are light red in colour with yellowish tinge.
- These bricks have rough surface and edges are not proper.

#### Uses:

- Third class bricks are used in temporary structures and unimportant structures as they have poor quality.
- These bricks are not suitable for rainy areas.

### 4) Fourth class bricks

- These are the over burnt bricks and due to this these bricks are brittle.
- They are dark in colour.
- The shape is irregular.
- These bricks are easily breakable and not suitable for construction.

#### Uses:

- Fourth class bricks have poor quality and not used for construction.

- They are used as aggregates for concrete in foundation, floors and roads.

## **CEMENT CONCRETE**

- Cement concrete, commonly known as concrete, is a composite material composed of cement, coarse aggregates, fine aggregates and water in suitable proportions.
- When these ingredients are mixed together, they form a strong and durable material that is widely used for construction.
- The chemical interaction between cement and water binds the aggregates in to a solid mass.
- Fresh concrete will be plastic, so that it can be moulded to any desired shape in the moulds.
- The strength of the concrete increases with age.
- Usually setting completes with in a duration of 10 hours, while about 90% of hardening is completed by 28 days.
- Concrete is used extensively in construction for buildings, bridges, roads, dams, and many other structures due to its versatility, strength, durability, and relatively low cost compared to other building materials.
- Concrete structures have become very common in civil engineering construction.
- Concrete has established as a universal building material because of the following reasons
  - High compressive strength
  - Adoptability to take any form and shape
  - Resistant to fire and corrosion with negligible maintenance cost.

## **FUNCTIONS OF INGREDIENTS OF CONCRETE**

- The properties and quality of cement concrete are influenced by the properties of it's ingredients and quality control maintained during making and curing.
- The ingredients of concrete are
  - a) Cement
  - b) Aggregates
  - c) Water

### **a) Cement**

- Cement is the binding material that holds the concrete together.
- Portland cement is the most common type used in concrete production.
- It reacts with water to form a paste that binds the aggregates together and hardens over time through a process called hydration.

### **b) Aggregates**

- Around 75% volume of concrete is occupied by the aggregates.
- Hence, the structural behavior of concrete is significantly influenced by the type of aggregates used.

- The aggregates used for the concrete should be durable, strong, hard, chemically inert and well graded.
- For making concrete both fine aggregates and coarse aggregates are required.

## (i) Fine aggregates

- Usually, sand is used as fine aggregate
- Fine aggregates help fill the voids between the larger particles, such as gravel or crushed stone.
- They contribute to the workability and density of the concrete mixture.

## (ii) Coarse aggregates

- These are larger particles of gravel or crushed stone that provide strength to the concrete.
- They also help to reduce the amount of cement needed, which can make the concrete more economical.

## c) Water

- Water used for making concrete should be clean.
- Water is essential for the hydration process of cement.
- The right amount of water is crucial for achieving the desired consistency and strength in concrete.
- Too much water can weaken the mixture, while too little water can make it difficult to work with.

## REINFORCED CEMENT CONCRETE

- Concrete is very strong in compression but very weak in tension.
- It's low tensile strength is compensated by introducing steel reinforcement in the tension zone.
- Thus, the concrete is strengthened by steel and the resultant composite mass is known as Reinforced Cement Concrete.

## CURING OF CONCRETE

- Curing may be defined as the process of maintaining satisfactory moisture and temperature conditions for freshly placed concrete for some specified time for proper hardening of concrete.
- Curing in the early ages of concrete is more important.
- Curing for 14 days is very important, better to continue it for 7 to 14 days more.
- If curing is not done properly, the strength of concrete reduces.
- Cracks will develop due to shrinkage.
- The durability of concrete structure reduces.

The following curing methods are employed

(a) Spraying of water

(b) Covering the surface with wet gunny bags, straw etc.

- (c) Ponding
- (d) Steam curing and
- (e) Application of curing compounds.

**(a) Spraying of water**

Walls, columns, plastered surfaces are cured by sprinkling water.

**(b) Wet covering the surface**

Columns and other vertical surfaces may be cured by covering the surfaces with wet gunny bags or straw.

**(c) Ponding**

The horizontal surfaces like slab and floors are cured by stagnating the water to a height of 25 to 50 mm by providing temporary small hunds with mortar.

**(d) Steam curing**

- In the manufacture of pre-fabricated concrete units steam is passed over the units kept in closed chambers.
- It accelerates curing process, resulting into the reduction of curing period.

**(e) Application of curing compounds**

- Compounds like calcium chloride may be applied on the curing surface.
- The compound shows affinity to the moisture and retains it on the surface.
- It keeps the concrete surface wet for a long time.

**PROPERTIES OF CONCRETE**

- Concrete has completely different properties when it is the plastic stage and when hardened.
- Concrete in the plastic stage is also known as fresh concrete or green concrete.

**PROPERTIES OF FRESH CONCRETE**

The properties of fresh concrete or green concrete are

- 1) Workability
- 2) Segregation
- 3) Bleeding
- 4) Harshness

**1) Workability**

- Workability is an important property of concrete in it's stage.
- Workability in simple terms can be defined as "the ease with which the concrete cane be mixed, transported, placed and compacted".
- The workability of concrete has also been defined as the amount of work required to place concrete and to compact.



- The workability depends upon the quantity of water, grading, shape and the percentage of the aggregates present in the concrete.

Workability is measured by

- (a) The slump observed when the frustum of the standard cone filled with concrete is lifted and removed.
- (b) The compaction factor determined after allowing the concrete to fall through the compaction testing machine.
- (c) The time taken in seconds for the shape of the concrete to change from cone to cylinder when tested in Vee-Bee consistometer.

## **2) Segregation**

- Separation of coarse particles from the green concrete is called segregation.
- This may happen due to lack of sufficient quantity of finer particles in concrete or due to throwing of the concrete from greater heights at the time of placing the concrete.
- Because of the segregation, the cohesiveness of the concrete is lost.
- Ultimately it results in the loss of strength of hardened concrete.
- Hence utmost care is to be taken to avoid segregation.

## **3) Bleeding**

- This refers to the appearance of the water along with cement particles on the surface of the freshly laid concrete.
- This happens when there is excessive quantity of water in the mix or due to excessive compaction.
- Bleeding causes the formation of pores and renders the concrete weak.
- Bleeding can be avoided by suitably controlling the quantity of water in the concrete and by using finer grading of aggregates.

## **4) Harshness**

- Harshness is the resistance offered by concrete to its surface finish.
- Harshness is due to presence of lesser quantity of fine aggregates, lesser cement mortar and due to use of poorly graded aggregates.
- It may result due to insufficient quantity of water also. With harsh concrete it is difficult to get a smooth surface finish and concrete becomes porous.

## **PROPERTIES OF HARDENED CONCRETE**

The properties of hardened concrete are

- 1) Strength
- 2) Durability
- 3) Impermeability
- 4) Resistance to wear and tear



### 1) Strength

- Hardened concrete should have high compressive strength.
- Compressive strength is the most important parameter considered in the design of concrete structures.
- The characteristic strength of concrete is defined as the compressive strength of 150 mm size cubes after 28 days of curing below which not more than 5 per cent of the test results are expected to fail.
- The unit of stress used is  $\text{N/mm}^2$ .
- IS 456 grades the concrete based on its characteristic strength.
- Concrete grade is denoted by letter M (mix ) and one number which is the characteristic strength at 28 days in  $\text{N/mm}^2$

| Grade of Concrete                                     | M 5    | M 7.5 | M 10  | M 15  | M 20    | M 25  |
|-------------------------------------------------------|--------|-------|-------|-------|---------|-------|
| Characteristic strength at 28 days in $\text{N/mm}^2$ | 5      | 7.5   | 10    | 15    | 20      | 25    |
| Proportion of mix                                     | 1:5:10 | 1:4:8 | 1:3:6 | 1:2:4 | 1:1.5:3 | 1:1:2 |

- IS 456–2000 specifies minimum grade of M20 to be used for reinforced concrete works.
- Strength of concrete depends upon the amount of cement content, quality and grading of aggregates, water cement ratio, compaction and curing.
- Strength of concrete is gained in the initial stages.
- In 7 days the strength gained is as much as 60 to 65 per cent of 28 days strength.
- It is customary to assume the 28 days strength as the full strength of concrete.

### 2) Durability

- Concrete should be capable of withstanding the adverse effects of weathering such as wind, water etc.
- In addition, it should resist temperature variation, moisture variation, freezing and thawing.
- The period of existence of concrete without getting adversely affected by these forces is known as durability.

### 3) Impermeability

- Hardened concrete should be impermeable and watertight.
- This is the resistance of concrete to the flow of water through its pores.
- This property of concrete is considered in the construction of water tanks.

#### 4) Resistance to wear and tear

- Hardened concrete should be capable of withstanding abrasive action during its usage.
- When concrete is used as a flooring material or for construction of cement concrete roads, it should withstand abrasive action.

#### DESIRABLE PROPERTIES OF FRESH CONCRETE

Appropriate quality and quantity of cement, fine aggregate, coarse aggregate and water should be used so that the fresh concrete has the following properties:

- (a) Desired workability
- (b) No segregation in transporting and placing
- (c) No bleeding and
- (d) No harshness

#### DESIRABLE PROPERTIES OF HARDENED CONCRETE

Hardened concrete should have the following properties

- (a) required characteristic strength
- (b) minimum dimensional changes
- (c) good durability
- (d) impermeable
- (e) good resistance to wear and tear

#### USES OF CONCRETE

- It is used for mass concrete works such as dams and bridges.
- It is used for the construction of silos and bunkers.
- It is used for the construction of road pavements.
- It is used for making electric poles, railways sleepers.
- It is used as bed concrete below column footings, wall footings, on wall at supports to beams.
- As sill concrete
- Bridges, dams, retaining walls are R.C.C. structures in which concrete is the major ingredient.

#### STEEL

- Steel is an alloy of iron and carbon containing less than 2% carbon and 1% manganese and small amounts of silicon, phosphorus, Sulphur and oxygen.
- Steel is the world's most important engineering and construction material.
- The steel structure is low in weight, which has many advantages over concrete structures and other structures.
- Steel is employed as reinforcement in construction because it is strong in both tension and compression, while concrete is only strong in compression and weak in tension.

## CHARACTERISTICS OF STEEL

### 1) Strength

Steel is known for its high tensile strength, making it suitable for structural applications.

### 2) Durability

Resistant to corrosion or rust and wear, especially stainless steel.

### 3) Malleability and Ductility

Can be shaped, rolled, and formed into various structures and products.

### 4) Heat Resistance

Some types of steel can withstand high temperatures without losing strength or shape.

### 5) Recyclability

Steel is highly recyclable, reducing environmental impact and conserving resources.

## ADVANTAGES OF STEEL

Steel has many advantages in the construction industry, they are

- High strength
- High durability
- Resistance to corrosion
- Resistance to fire
- Steel members can be easily handled and transported
- Steel structures have long life than RCC structures.
- Properties of steel do not change with time

## DISADVANTAGES OF STEEL

- It is susceptible to corrosion.
- Requires frequent painting to prevent corrosion.
- Maintenance cost is high.
- Steel members are costly.

## PROPERTIES OF GOOD STEEL

- Good durability
- Good tensile strength
- Good yield strength
- Good thermal conductivity
- Good formability

## TYPES OF STEEL

The following three varieties of steel are extensively used

(a) Mild steel

- (b) High carbon steel
- (c) High tensile steel

## (a) Mild Steel

It contains a maximum of 0.25% carbon, 0.055% of Sulphur and 0.55% of phosphorus.

### Properties of Mild Steel

- It is malleable and ductile
- It is more elastic
- It can be magnetized permanently
- Its specific gravity is 7.8
- Its Young's modulus is  $2.1 \times 10^5 \text{ N/mm}^2$
- It can be welded easily
- It is equally strong in tension and in compression.

### Uses of Mild Steel

- Round bars are extensively used as reinforcement in R.C.C. works.
- Rolled sections like I, T, L, C, plates etc. are used to build steel columns, beams, trusses etc.
- Tubular sections are used as poles and members of trusses.
- Plain and corrugated mild steel are used as roofing materials.
- Mild steel sections are used in making parts of many machineries.

## b) High Carbon Steel

- The carbon contains in this steel is 0.7% to 1.5%.

### Properties of High Carbon Steel

- It is more tough and elastic compared to mild steel.
- Welding is difficult.
- It can be magnetized permanently.
- It is stronger in compression than in tension.
- It withstands shocks and vibrations better.

### Uses of High Carbon Steel

- It is used for making tools such as drills, files, chisels.
- Many machine parts are made with high carbon steel since it is capable of withstanding shocks and vibrations.

## c) High Tensile Steel

- It contains 0.8% carbon and 0.6% manganese.
- The strength of this steel is quite high.

## Properties of High Tensile Steel

- High Strength
- Lightweight
- Excellent Toughness
- Good Ductility
- High Yield Strength
- Corrosion Resistance
- Temperature Resistance

## Uses of High Tensile Steel

- High tensile steel wires are used in prestressed concrete works.
- It is used in bridge building, mining, crane construction, mechanical engineering, safety constructions etc.

## Various grades of steel

- 1) Mild steel bars (Fe250)
- 2) High yield strength deformed bars (Fe415)
- 3) High yield strength deformed bars (Fe500)

## TIMBER

- Timber refers to wood used for construction works.
- In fact the word timber is derived from an old English word 'Timbrian' which means 'to build'.
- The timber thus denotes wood which is suitable for building or carpentry or various other engineering purposes.
- Timber was used as building material even by primitive man.
- Many ancient temples, palaces and bridges built with timber can be seen even today.

The following three terms are to be noted in connection with the timber

- 1) Standing timber
- 2) Rough timber
- 3) Converted timber

### 1) Standing Timber

A tree that yields good wood for construction is called Standing Timber

### 2) Rough Timber

- After felling a tree, its branches are cut and its stem is roughly converted into pieces of suitable length, so that it can be transported to timber yard.
- This form of timber is known as rough timber.

### 3) Converted Timber

- By sawing, rough timber is converted into various commercial sizes like planks, battens, posts, beams etc.
- Such form of timber is known as converted timber.

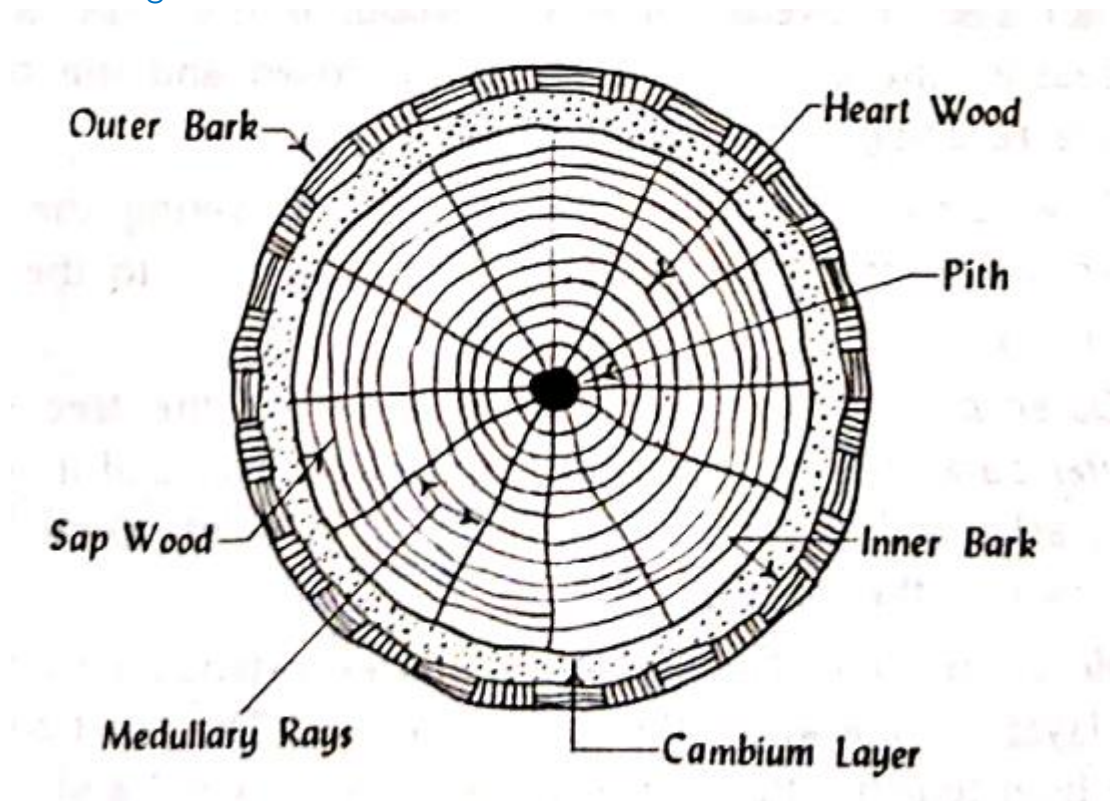
### Structure of a tree

From the visibility aspect, the structure of a tree can be divided into two categories, they are

- 1) Macrostructure
- 2) Microstructure

#### 1) Macrostructure

- The structure of wood visible to the naked eye is called macrostructure.
- The below figure shows the transverse section of the trunk of a tree.



The following components are visible to the naked eye

#### a) Pith

- It is the inner most part or core of the tree.
- It varies in size and shape for different types of trees.
- When the plant becomes old, the pith dies and becomes fibrous and dark.

#### b) Heart wood

- This is the portion surrounding pith.
- It is dark in colour and strong.
- This portion is useful for various engineering purpose.



- This is the dead part of wood.
- It consists of several annular rings.

**c) Sap wood**

- The outer annular rings between heartwood and cambium layer is known as sap wood.
- It is usually light in colour and weight.
- It indicates recent growth and contains sap.
- It takes active part in the growth of trees by allowing sap to move in upward direction.
- The annual rings of sap wood are less sharply divided than those of heartwood.
- The sap wood is also known as alburnum.

**d) Cambium layer**

- The thin layer of sap between sap wood and inner bark is known as cambium layer.
- It contains sap which is not yet converted into sap wood.
- If the bark is removed and cambium layer is exposed to atmosphere, cells cease to be active and tree dies.

**e) Inner bark**

- It is a inner skin of tree protecting the cambium layer.
- It gives protection to cambium layer.

**f) Outer Bark**

- It is the outer skin of the tree and consists of wood fibers.
- Sometimes it contains fissures and cracks.

**g) Medullary rays**

- The thin radial fibres extending from pith cambium layer are known as medullary rays.
- They hold annular rings together.
- In some of trees they are broken and some other they may not be prominent.

**2) Microstructure**

- The structure of the wood apparent only at great magnifications is called microstructure.
- When studied under a microscope, it becomes evident that wood consists of living and dead cells of various sizes and shapes.

**Properties of Good Timber**

The properties of good timber are

## 1) Colour

It should be uniform.

## 2) Oduor

It should be pleasant when cut freshly.

## 3) Soundness

A clear ringing sound when struck indicates the timber is good.

## 4) Texture

Texture of good timber is fine and even.

## 5) Grains

In good timber grains are close.

## 6) Density

Higher the density stronger is the timber.

## 7) Hardness

Harder timbers are strong and durable.

## 8) Warping

Good timber do not warp under changing environmental conditions.

## 9) Toughness

Timber should be capable of resisting shock loads.

## 10) Abrasion

- Good timber do not deteriorate due to wear.
- This property should be looked into, if timber is to be used for flooring.

## 11) Strength

Timber should have high strength in bending, shear and direct compression.

## 12) Modulus of Elasticity

Timber with higher modulus of elasticity are preferred in construction.

## 13) Fire resistance

A good timber should have high resistance to fire.

## 14) Permeability

Good timber has low water permeability.



**15) Workability**

- Timber should be easily workable.
- It should not clog the saw.

**16) Durability**

Good timber is one which is capable of resisting the action of fungi and insect's attack

**17) Defects**

Good timber is free from defects like dead knots, shakes and cracks.

**Seasoning of timber**

- Seasoning of Timber is the process of removing moisture content from freshly cut timber.
- Proper seasoning is crucial to enhance the quality and durability of timber products.
- Seasoning of timber is a critical step in preparing timber for various uses, ensuring it meets quality standards and performs well over time.

**Methods of seasoning of timber**

There are two methods of seasoning of timber

- 1) Natural seasoning
- 2) Artificial seasoning

**1) Natural seasoning**

It may be air seasoning or water seasoning

**a) Air seasoning**

- This is a traditional method where timber is stacked in a well-ventilated area and exposed to natural air circulation.
- Air seasoning allows moisture to gradually evaporate from the wood, reducing its moisture content over time.
- The duration of air drying depends on factors like the type of wood, thickness of the timber, humidity levels, and prevailing weather conditions.
- A well-seasoned timber contains only 15% moisture.
- This is a slow but a good process of seasoning.

**b) Water seasoning**

- Water seasoning is the process in which timber is immersed in water flow which helps to remove the sap present in the timber.
- After a period of 2 to 4 weeks the timber is taken out.
- During this period sap contained in the timber is washed out to a great extent.
- After that the timber is allowed to dry.

## 2) Artificial Seasoning

- In this method timber is seasoned in a chamber with regulated heat, controlled humidity and proper air circulation.
- Seasoning can be completed in 4 to 5 days only.
- The different methods of seasoning are
  - (a) Boiling
  - (b) Kiln seasoning
  - (c) Chemical seasoning
  - (d) Electrical seasoning.

### (a) Boiling

- In this method timber is immersed in water and then water is boiled for 3 to 4 hours.
- Then it is dried slowly.
- Instead of boiling water hot steam may be circulated on timber.
- The boiling or steaming process develops the strength and elasticity of timber
- The process of seasoning is fast, but costly.

### b) Kiln seasoning

- Kiln is an airtight chamber.
- Timber to be seasoned is placed inside it.
- Then fully saturated air with a temperature  $35^{\circ}\text{C}$  to  $38^{\circ}\text{C}$  is forced in the kiln.
- The heat gradually reaches inside timber.
- Then relative humidity is gradually reduced and temperature is increased, and maintained till desired degree of moisture content is achieved.
- The kiln used may be stationary or progressive.
- In progressive kiln the carriages carrying timber travel from one end of kiln to other end gradually.
- The hot air is supplied from the discharging end so that temperature increase is gradual from charging end to discharging end.
- This method is used for seasoning on a larger scale.

### (c) Chemical Seasoning

- This is also known as salt seasoning.
- In this method, the timber is immersed in a solution of suitable salt.
- The salt solution used has the tendency to absorb water from the timber.
- So, the moisture content is removed and then timber is allowed to drying.
- The inorganic salts as sodium chloride, ferrous sulphate, sodium sulphate, aluminium sulphate, calcium acetate, sodium phosphate, barium chloride, magnesium sulphate etc are used for chemical seasoning of timber.
- The preliminary treatment by chemical seasoning ensures uniform seasoning of outer and inner parts of timber.

**(d) Electrical Seasoning**

- In this method high frequency alternate electric current is passed through timber.
- Resistance to electric current is low when moisture content in timber is high.
- As moisture content reduces the resistance reduces.
- Measure of resistance can be used to stop seasoning at appropriate level.
- It is a very fast method of seasoning.
- However, it is costly process.
- This technique has been tried in some plywood industries but not in seasoning of timber on mass scale.

**9. Practice Quiz****1. Civil engineers contribute the following to the welfare of the society**

- a) Planning of towns
- b) Water supply
- c) Transportation

**d) All of the above****2. \_\_\_\_\_ engineering deals with the structural design and analysis of buildings, bridges etc.**

- a) Structural**
- b) Geotechnical
- c) Transportation
- d) Environmental

**3. \_\_\_\_\_ engineering deals with the behavior of soils and rocks and design of foundation.**

- a) Structural
- b) Geotechnical**
- c) Transportation
- d) Environmental

**4. The branch of engineering that deals with the construction of roads, bridges etc is called \_\_\_\_\_**

- a) Structural
- b) Geotechnical
- c) Transportation**
- d) Environmental

**5. The branch of engineering that deals with the purification of drinking water is called**

- a) Structural
- b) Geotechnical
- c) Transportation
- d) Environmental**



**6. Cement was invented in the year**

- a) 1624
- b) 1724
- c) 1824**
- d) 1924

**7. The fineness of the cement can be found out by sieve analysis by using IS sieve number**

- a) 60 micron
- b) 70 micron
- c) 80 micron
- d) 90 micron**

**8. According to Indian standards, the size of the brick is (in cm)**

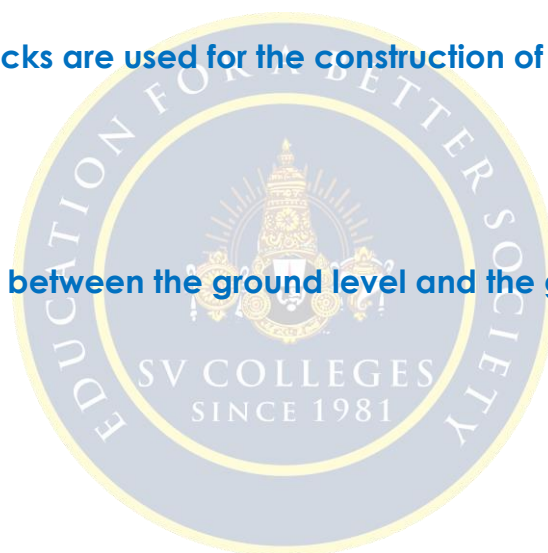
- a) 19 X 9 X 9**
- b) 20 X 9 X 9
- c) 19 X 10 X 10
- d) 20 X 10 X 10

**9. \_\_\_\_\_ class bricks are used for the construction of load bearing walls.**

- a) First**
- b) Second
- c) Third
- d) Fourth

**10. The portion of the wall between the ground level and the ground floor is called**

- a) Plinth**
- b) Lintel
- c) Sill
- d) Chejjas



## 10. Assignments

| S.No | Question                                                                                            | BL | CO |
|------|-----------------------------------------------------------------------------------------------------|----|----|
| 1    | Briefly explain the scope of different fields in Civil Engineering.                                 | 2  | 1  |
| 2    | What is cement? State chemical and physical properties of cement.                                   | 2  | 1  |
| 3    | Explain briefly the classification of coarse aggregates.                                            | 2  | 1  |
| 4    | State the properties of fresh concrete and hardened concrete.                                       | 2  | 1  |
| 5    | Describe the properties of good bricks.                                                             | 2  | 1  |
| 6    | What is meant by seasoning of timber? Briefly explain the different methods of seasoning of timber. | 2  | 1  |



## 11. Part A- Question &amp; Answers

| S.No | Question & Answers                                                                                                                                                                                                                                                                        | BL | CO |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|
| 1    | <b>List various disciplines of Civil Engineering.</b><br><b>Ans.</b> 1) Structural Engineering<br>2) Geotechnical Engineering<br>3) Transportation Engineering<br>4) Hydraulics & Water Resources Engineering<br>5) Environmental Engineering                                             | 1  | 1  |
| 2    | <b>What are building materials?</b><br><b>Ans.</b> The materials used in the construction of engineering structures such as buildings, bridges and roads are called construction materials or building materials or engineering materials.                                                | 1  | 1  |
| 3    | <b>What is cement?</b><br><b>Ans.</b> Cement is commonly used binding material in construction.                                                                                                                                                                                           | 1  | 1  |
| 4    | <b>What are the raw materials required for manufacture of Portland cement?</b><br><b>Ans.</b> The cement is obtained by burning a mixture of calcareous (calcium) and argillaceous (clay) material at a very high temperature and then grinding the clinker so produced to a fine powder. | 1  | 1  |
| 5    | <b>What is meant by hydration of cement?</b><br><b>Ans.</b> Hydration is the chemical reaction between cement and water by which Portland cement becomes a firm, hardened mass.                                                                                                           | 1  | 1  |
| 6    | <b>List the properties of good sand.</b><br><b>Ans.</b> 1) It should be clean and coarse.<br>2) It should be free from any organic matter or vegetable matter.<br>3) It should be chemically inert.<br>4) It should contain sharp, angular, coarse and durable grains.                    | 1  | 1  |
| 7    | <b>What are the different uses of bricks?</b><br><b>Ans.</b> 1) Bricks are extensively used as a leading material of construction.<br>2) Bricks are extensively used for construction of load-bearing walls and partition walls.                                                          | 2  | 1  |
| 8    | <b>What is meant by segregation?</b><br><b>Ans.</b> Separation of coarse particles from the green concrete is called segregation.                                                                                                                                                         | 2  | 1  |
| 9    | <b>What is meant by bleeding?</b><br><b>Ans.</b> This refers to the appearance of the water along with cement particles on the surface of the freshly laid concrete.                                                                                                                      | 2  | 1  |
| 10   | <b>What is meant by converted timber?</b><br><b>Ans.</b> By sawing, rough timber is converted into various commercial sizes like planks, battens, posts, beams etc. Such form of timber is known as converted timber.                                                                     | 2  | 1  |

## 12. Part B- Questions

| S.No | Question                                                                                                                | BL | CO |
|------|-------------------------------------------------------------------------------------------------------------------------|----|----|
| 1    | Briefly explain the scope of structural engineering and transportation engineering specialization of Civil Engineering. | 2  | 1  |
| 2    | What is meant by sand? What are the natural sources of sand?                                                            | 1  | 1  |
| 3    | What do you understand by the term brick? Explain different types of bricks.                                            | 2  | 1  |
| 4    | Explain the classification of bricks.                                                                                   | 2  | 1  |
| 5    | Draw and explain cross-section of an exogeneous tree.                                                                   | 2  | 1  |

## 13. Supportive Online Certification Courses

1. Introduction to Civil Engineering Profession By Prof. Subhadeep Banerjee, conducted by IIT Madras – 8 weeks

## 14. Prescribed Text Books &amp; Reference Books

## Text Book

1. Basic Civil Engineering, M. S. Palanisamy, Tata Mcgraw Hill publications (India) Pvt. Ltd. Fourth Edition.
2. Introduction to Civil Engineering, S.S. Bhavikatti, New Age International Publishers. 2022. First Edition.

## References:

1. Hydrology and Water Resources Engineering, Santosh Kumar Garg, Khanna Publishers, Delhi. 2016
2. Highway Engineering, S.K.Khanna, C.E.G. Justo and Veeraraghavan, Nemchand and Brothers Publications 2019. 10th Edition.